

NANYANG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATIONS
Higher 1

CANDIDATE
NAME

CLASS

BIOLOGY

8875/02

Paper 2 Core Paper

14 September 2011

Additional Materials: Answer Paper

2 hours

READ THESE INSTRUCTIONS FIRST

Write your name and CT on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use soft pencil for any diagrams, graphs or rough working.
Do no use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer any **one** question.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	
2	
3	
4	
Section B	
Total	

This document consists of **10** printed pages and **no** blank pages.

[Turn over

Section A

Answer **all** the questions in this section.

- 1 Lipase is an enzyme that catalyses the hydrolysis of triglycerides. It is a soluble, globular protein. The function of an enzyme depends upon the precise structure of its tertiary structure. Fig. 1.1 represents the structure of lipase. The black, solid strips represent the disulphide bonds which help to stabilize the tertiary structure.

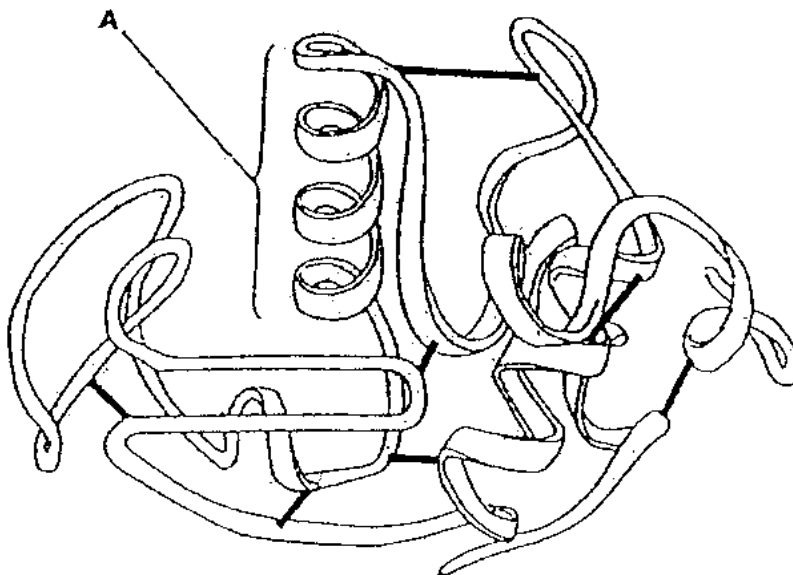


Fig 1.1

- (a) With reference to **Fig 1.1** above,
- (i) Name **two** other types of bonds that help to stabilize the tertiary structure of lipase

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[2]

- (ii) Region **A** is a motif commonly found in lipase. Describe how it is formed from amino acids.

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[3]

$$\begin{array}{c}
 \text{(Saturated) Fatty Acid}_1 - \overset{\overset{\text{O}}{\parallel}}{\text{C}} - \text{O} - \text{CH}_2 \\
 | \\
 \text{(Unsaturated) Fatty Acid}_2 - \underset{\underset{\text{O}}{\parallel}}{\text{C}} - \text{O} - \text{CH} \\
 | \qquad \qquad \qquad \parallel \qquad \qquad \qquad \text{O} \qquad \qquad \qquad + \\
 \text{CH}_2 - \text{O} - \text{P} - \text{O} - \text{CH}_2 - \text{CH} \quad \text{NH}_3 \\
 | \qquad \qquad \qquad | \qquad \qquad \qquad / \\
 \text{O}^- \qquad \qquad \qquad \text{COO}^-
 \end{array}$$

Fig 1.2

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(c) Describe how the structure of the biomolecule **A** contributes to their function in membranes.

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[Total: 10]

- 2 A monkey homozygous for a mutation on the insulin gene suffers from Type 1 diabetes. An experiment utilizing genetic engineering techniques was conducted to obtain genetically modified stem cells in order to treat the disease.

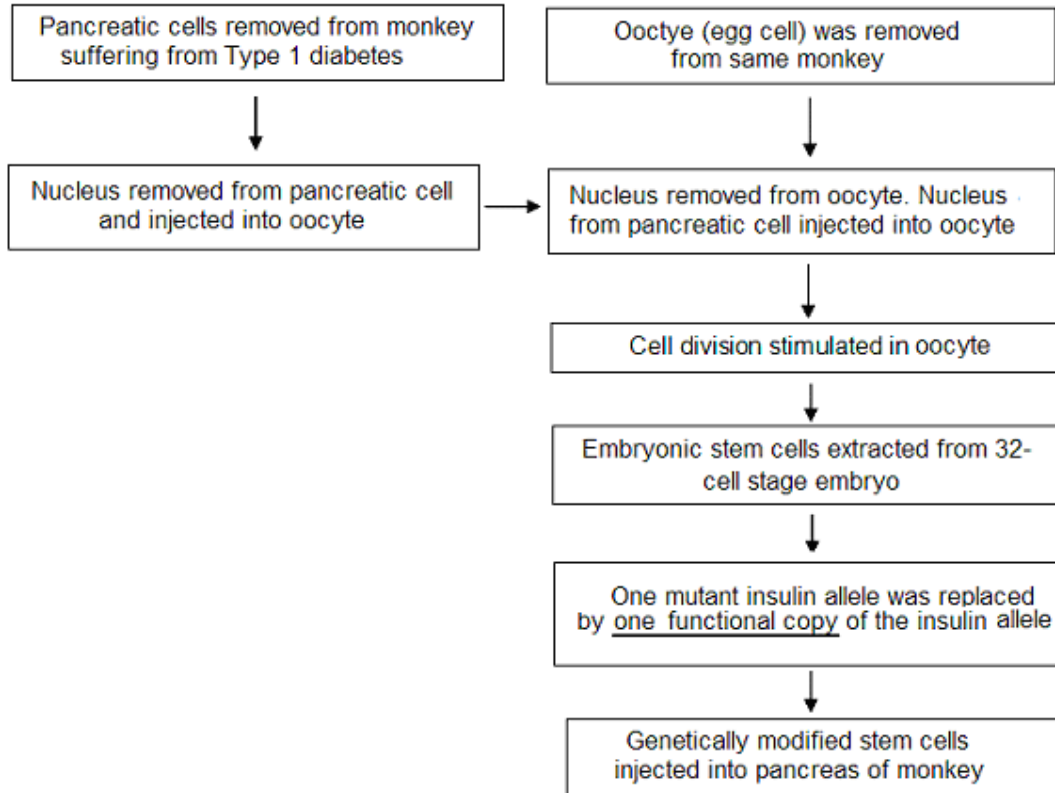


Fig. 2.1

- (a) Describe the features of embryonic stem cells.

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[2]

- (b) Describe the role of embryonic stem cells in a normally developing embryo.

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[2]

- (c) Suggest why it is sufficient to insert a single functional allele into the stem cell.

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[1]

One method of treating diabetes in humans involves direct injection of insulin. Insulin is a peptide hormone that is involved in controlling blood sugar levels and is one of the first recombinant proteins from humans to be expressed in *Escherichia coli* for medical applications.

Fig 2.2 shows the proinsulin molecule produced by *E.coli*.

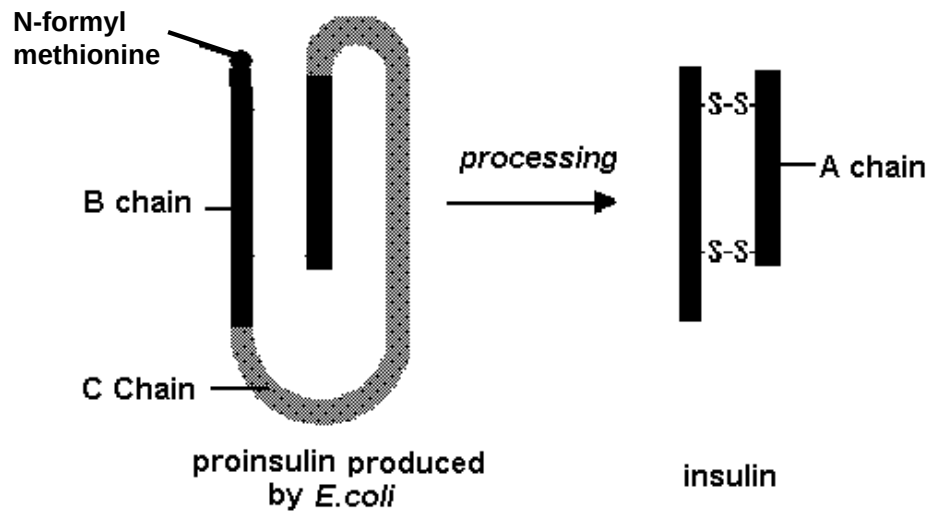


Fig. 2.2

- (d) With reference to **Fig. 2.2**, outline the steps that must be undertaken during the processing phase in order to obtain insulin.

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[3]

- (e) Other than the problems visible in **Fig 2.2**, state one other problem associated with using bacteria to produce eukaryotic proteins and suggest how it can be overcome.

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[2]

[Total: 10]

- 3** Coat colour in cats is determined by a sex-linked gene with two alleles, black and ginger. When black cats are mated with ginger cats, the female progeny are always tortoiseshell, their coats showing a mottling of small black and ginger patches, while the male progeny have the same coat colour as their mothers.
- (a)** Using suitable symbols, construct genetic diagrams to explain these results.

- (b) In a litter of kittens, half the number of females were tortoiseshell and the other half were black; half the number of males were black and the other half were ginger. State and explain what the colours of the parental male and female were.

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[4]

[Total: 10]

4 Fig 4.1 represents the main sequence of events in oxidative phosphorylation.

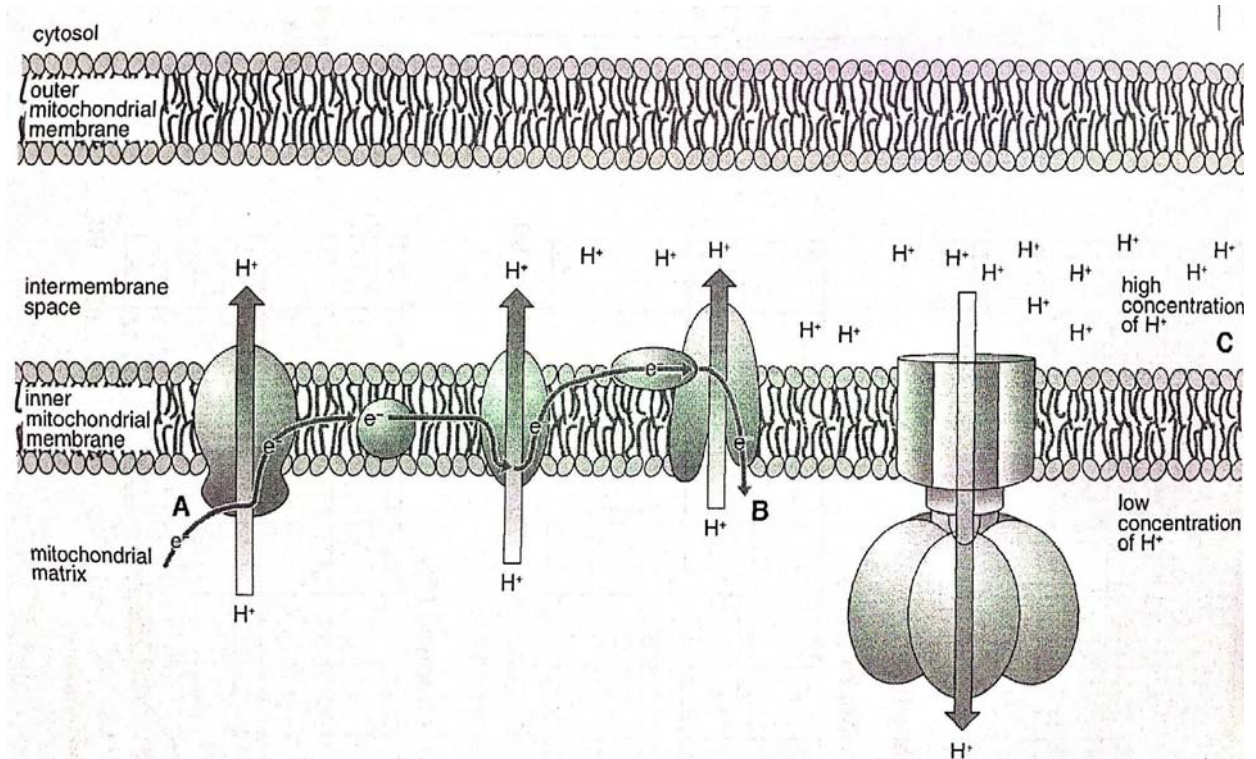


Fig 4.1

(a) Describe the source of electrons at A.

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[2]

(b) Describe the fate of the electrons at B

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[2]

(c) Describe the role of the high concentration of protons at C

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[2]

- (d) Explain how the flow of electrons from **A** to **B** results in the high concentration of protons at **C**.

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[2]

- (e) List three ways in which photophosphorylation differs from oxidative phosphorylation.

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[2]

[Total: 10]

Section B

Answer **one** question.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 5** (a) Outline the structure and functions of membranes within animal cells. [8]
- (b) Explain the need for reduction division (meiosis) and how it can lead to variation. [6]
- (c) Explain how uncontrolled cell division can result in cancer, and identify the factors which can increase the chances of cancerous growth. [6]
- 6** (a) Describe the structural features of two named nucleic acids involved in protein synthesis. [6]
- (b) Compare the two processes that have to take place in order for protein synthesis to occur. [8]
- (c) Briefly describe the process of DNA replication. [6]